

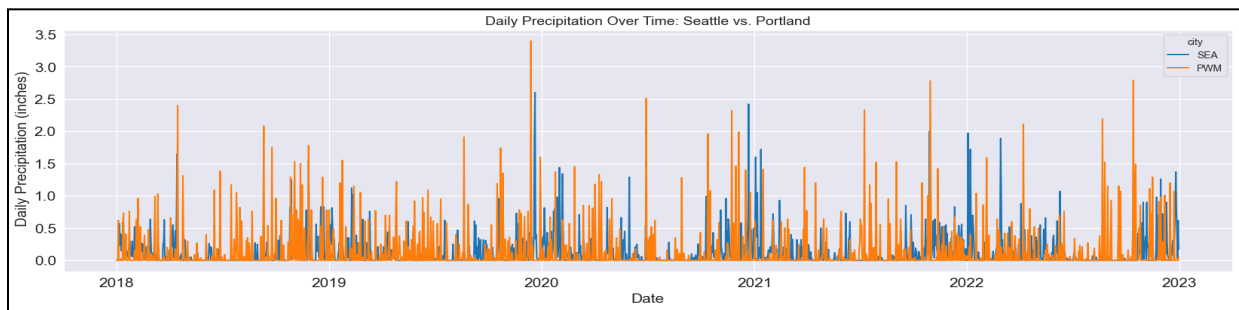
Which City Gets More Rain?

Seattle, Washington vs. Portland, Maine

Introduction:

Seattle, Washington has a reputation for being a very rainy city. To see if this reputation is exaggerated, I chose to compare it to a city on the opposite coast, Portland, Maine. Using historical daily precipitation data I will aim to answer the question: Does it rain more in Seattle, WA, or Portland, ME?

The goal of this analysis was to compare how much it rains and how often it rains in both cities. To do this, daily precipitation from records from 2018 to 2022 were accessed from the National Oceanic and Atmospheric Administration (NOAA).



Methods:

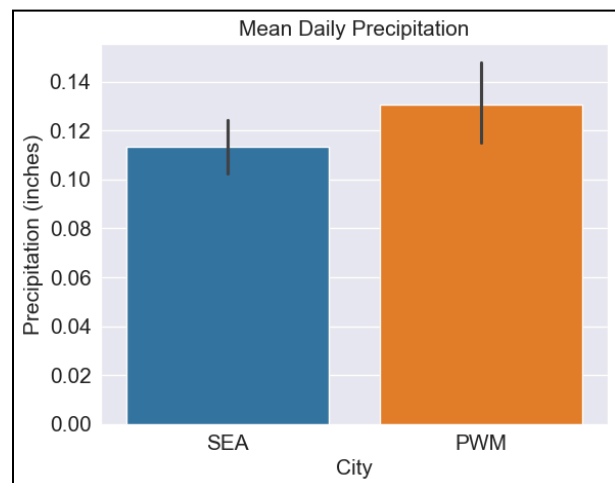
Data from both datasets included the date, city, amount of daily precipitation (rain) in inches and were combined into one in order to process and analyze. Missing data was replaced by using the average precipitation for that day of the year in order to preserve seasonal patterns.

Descriptive statistics were used to summarize how much it rained in each city and any seasonal patterns that were present. Two types of statistical testing were used to compare the average daily precipitation between the two cities and the proportion of days that had rain. Finally, visualizations were created from the data to give a quick visual representation of the findings.

Results:

Average Daily Rain:

Portland averaged about 0.13 inches of rain a day, while Seattle averaged about 0.11 inches. This small difference over the five year



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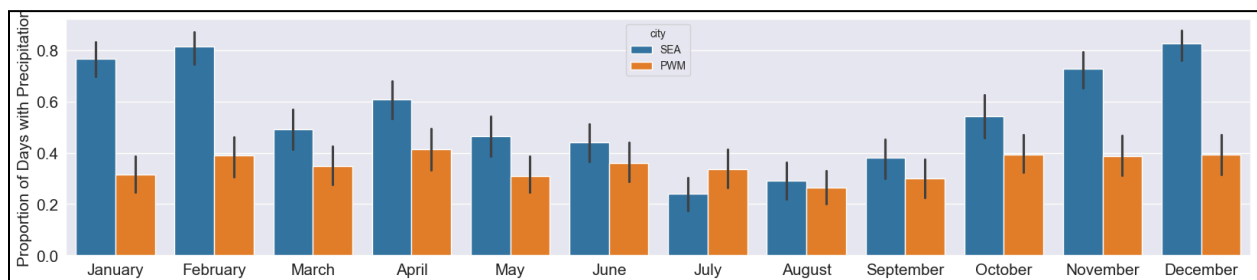
Seattle, Washington vs. Portland, Maine

period adds up to Portland having roughly 15% more rain than Seattle (238 inches vs 206 inches).

When looked at by months, November to January were Seattle's peak rainy months dropping to its driest from June to August. Overall, Portland's rain was more evenly distributed throughout the year, though it gets heavier single day rain (max = 3.4 inches vs. 2.6 inches).

Frequency of Rainy Days:

Seattle recorded rain on about 55% of the days within the dataset, compared to about 35% in Portland. The 20% difference can likely be attributed to the eight months (January to May, and October to December) that Seattle had significantly more rainy days. Statistical testing done showed that those eight months were significant, but June to September were similarly dry in both cities.



Conclusion:

The analysis compared rain patterns between Seattle, Washington and Portland, Maine to test whether one city does actually rain more than the other. The analysis shows that across the years within the dataset it rained about 55% of the days, compared to Portland's 35%. However, Portland received roughly 238 inches of rain which is about 15% more than Seattle's 206 inches.

Seattle's reputation as a rainy city, may or may not be exaggerated depending on the perspective. If the question being asked is if Seattle gets rain more, then it is more rainy than Portland. However, if it's being asked from the perspective of the amount of rain, then Portland is more rainy.